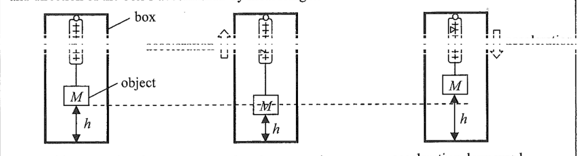
ch6 Force

9 item(s)

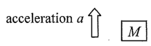
4. Read the following description about **accelerometers** and answer the questions that follow.

An accelerometer is a device for measuring acceleration. The following example illustrates the principle of a simple accelerometer. An object of mass M is suspended by a spring balance inside a box. If the box is at rest without acceleration, the object is h above the bottom of the box (Figure 4.1). When the box accelerates upward, h decreases (Figure 4.2). Likewise, when the box accelerates downward, h increases (Figure 4.3). Since it is known that the tension of the spring balance is directly proportional to its extension, we can therefore determine the magnitude and direction of the box's acceleration by measuring h.



at rest without acceleration Figure 4.1 | accelerating upward Figure 4.2 | accelerating downward Figure 4.3

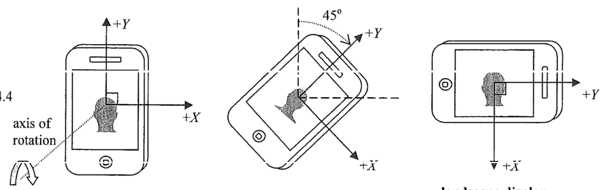
(a) Draw a labelled free-body diagram in the space provided below showing the forces acting on the object when the box accelerates upward with acceleration a. Explain why h decreases in this case. (4 marks)



(b) The scale of the spring balance is calibrated such that the pointer moves 1 cm for a change of 2 N of force. The weight of the object is 5 N. If h decreases by 0.5 cm compared to the situation in Figure 4.1, what is the reading of the balance ? Hence find the magnitude of the corresponding acceleration of the box. (acceleration due to gravity $g = 9.81\ \mathrm{m\ s^{-2}}$) (3 marks)

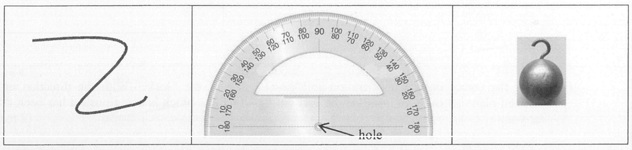
Electronic accelerometers employing similar principles are widely used in smart phones. The phone's orientation is detected by a set of built-in accelerometers each detecting the acceleration due to gravity along mutually perpendicular axes of the phone. A phone in vertical orientation is shown on the left of Figure 4.4, its accelerometer along the Y axis would be sensing the acceleration due to gravity, denoted by $a_Y = -g$. When the phone is rotated about a horizontal axis perpendicular to both X and Y axes by more than 45°, 'portrait display' would change to 'landscape display' as shown on the right of Figure 4.4.

Figure 4.4



(c) What would the kind of display be if the phone is rotated clockwise until the acceleration a_Y sensed by the accelerometer along the rotated Y axis is $-0.5\ g$? Explain. (2 marks)

5. You are given a long light string, a protractor and a metal ball with a hook.



Suppose you are inside a train which is at rest initially and later it travels along a straight horizontal track with constant acceleration. With the aid of a diagram, describe how to measure the acceleration of the train. Show your working including mathematical derivation. (6 marks)

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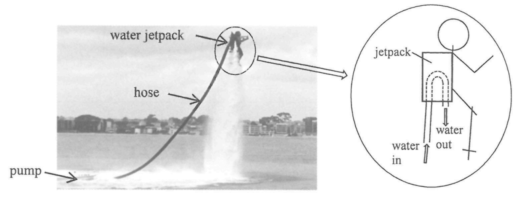
4.

Figure 4.1



A person wears a water jetpack which enables him to stay 'afloat' in equilibrium in the air as shown in Figure 4.1. A pump on the sea surface continuously pumps water to the jetpack via a hose and the water is then ejected downwards.

(a) Referring to Figure 4.1, water enters the U-shape hose inside the jetpack with a certain speed and is then ejected out vertically downwards. Use Newton's law(s) of motion to explain why a lifting force acting on the person is produced. (2 marks)

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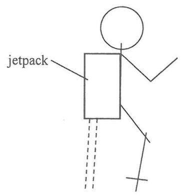
(b) Draw and label all the forces acting on the person wearing the jetpack as a whole in the free-body diagram below. Neglect the pulling force due to the hose connected to the jetpack. (1 mark)



(c) Suppose that water enters the jetpack with a speed of 10 m s^{-1} vertically upwards and is then ejected out at the same speed vertically downwards. ($g = 9.81 \text{ m s}^{-2}$)

(i) **Just by considering the change of momentum of the water**, estimate how much water, in kg, has to be ejected per second to provide a lifting force of 1000 N needed. (2 marks)

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(ii) Water is pumped to the water jetpack at a height of 7.5 m above sea surface and then ejected from it. By considering the gain in mechanical energy of the water, estimate the minimum output power of the pump. (2 marks)

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(d) The person changes to staying 'afloat' in equilibrium at a higher position. If the speed by which water enters and is ejected from the jetpack remains the same, would the amount of water ejected per second be greater than, equal to or smaller than the result found in (c)(i)? Explain. (Neglect the weight of the hose.) (2 marks)

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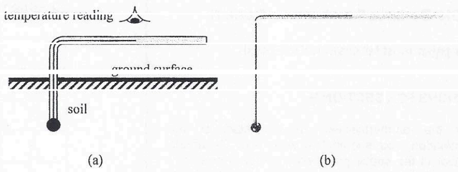
Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. Read the following passage about **soil thermometer** and answer the questions that follow.

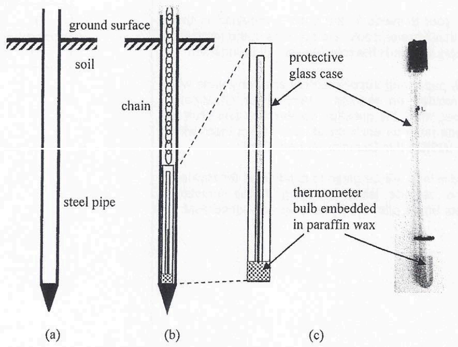
The temperature of soil changes with depth, and this information is important to farmers and scientists. To measure soil temperatures at depths close to the ground surface, the bulb of a thermometer is buried in the soil. The stem of the thermometer is bent 90° for easy reading. Figure 1.1a is a schematic diagram and Figure 1.1b shows a photo of a soil thermometer.

Figure 1.1



For depths greater than 30 cm, a steel pipe is driven into the soil (Figure 1.2a); and a liquid-in-glass thermometer with a protective glass case is lowered into the steel pipe (Figure 1.2b). The bulb of the thermometer is embedded in paraffin wax (Figure 1.2c). To read the temperature, the thermometer is lifted out of the steel pipe by pulling the chain.

Figure 1.2



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(a) As shown in Figure 1.1b, the bulb of the soil thermometer is very large compared to those of common thermometers. Suggest a reason for this design. (1 mark)

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(b) On a certain morning, the air temperature is 15°C. An observer takes a measurement of the soil temperature at 1 m deep. The thermometer reading is 20°C. It is given that the mass of the paraffin wax enclosing the thermometer bulb is 0.015 kg, and the specific heat capacity of paraffin wax is 2.9×10^3 J kg^{-1} °C^{-1}.

(i) Calculate the energy loss of the paraffin wax as it cools down to the air temperature. (2 marks)

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(ii) It is known that the paraffin wax enclosing the bulb of the thermometer gains or loses energy at a constant rate of 0.5 J s^{-1}, estimate the time taken for the paraffin wax to reach the air temperature after the thermometer is lifted out of the soil. (2 marks)

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(iii) If there is no paraffin wax enclosing the bulb of the thermometer, explain how the thermometer reading as recorded by the observer is affected. (2 marks)

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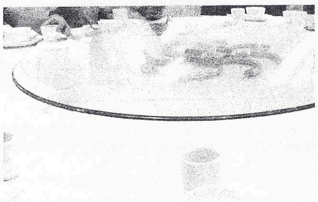
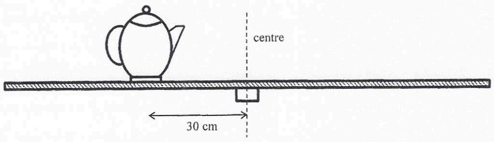
*5. The photo shows a turntable commonly used in restaurants.



A teapot of mass 1 kg is put 30 cm from the centre of a horizontal turntable, Figure 5.1 shows the side view. When the turntable is rotating, the teapot remains at the same position on the turntable.

Figure 5.1



(a) On Figure 5.1, draw and label all the forces acting on the teapot when the turntable is rotating. (2 marks)

(b) Taking the teapot as a point mass, estimate the net force acting on the teapot when the turntable is rotating at a rate of 0.5 revolutions per second. (3 marks)

(c) The turntable is suddenly stopped and the teapot slips. The turntable is rotating at a rate of 0.5 revolutions per second just before it stops, and the frictional force acting on the teapot is 10 N when it is slipping. Determine the distance travelled by the teapot after the turntable stops. (3 marks)

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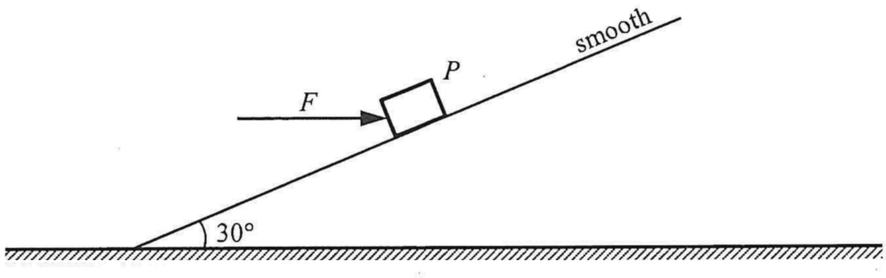
5.

Figure 5.1



(a) A block P of mass 10 kg is kept stationary on a smooth incline by a horizontal force F as shown in Figure 5.1. The incline makes an angle of 30° with the horizontal. ($g = 9.81\ \mathrm{m\ s^{-2}}$)

(i) On Figure 5.1, indicate and label all other forces acting on P. (2 marks)

(ii) Find the magnitudes of the force F and the force exerted by the block on the incline respectively. (3 marks)

(b) Now F is removed and neglect air resistance.

(i) What is the magnitude of the acceleration of the block ? (1 mark)

(ii) Explain whether the force exerted by the block on the incline would increase, decrease or remain unchanged when compared with that in (a)(ii). (2 marks)

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3. Figure 3.1 shows a quadcopter with four propellers.

Figure 3.1

When the four propellers are operating to propel air streams vertically downwards, the quadcopter can hover in the air in a fixed position. ($g = 9.81$ m s^{-2})

(a) Referring to Newton's laws of motion, explain why the quadcopter can hover in the air. (2 marks)

Given: mass of the quadcopter = 1.38 kg
total area swept by the **four** propellers = 0.284 m^2
density of air = 1.20 kg m^{-3}

(b) Suppose the speed of the air streams produced is v.

(i) Considering the total volume of air being propelled downwards in 1 second, express the mass of air m_a propelled by the quadcopter per second in terms of v. (2 marks)

(ii) Hence, find the speed v that enables the quadcopter to hover. (2 marks)

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(c) The quadcopter can be adjusted to tilt at an angle θ to the vertical as shown in Figure 3.2(a) and fly along a horizontal circular path of radius r (Figure 3.2(b)). Neglect the size of the quadcopter and air resistance in your calculation.

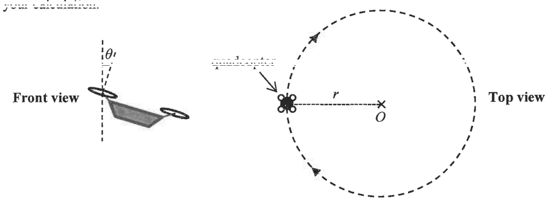

Figure 3.2(a)

Figure 3.2(b)

(i) On Figure 3.2(a), indicate and label the force(s) acting on the quadcopter. (2 marks)

*(ii) Find the centripetal force required for the quadcopter to follow such a circular path with a radius of 50 m and a speed of 15 m s^{-1}. (2 marks)

(iii) Hence, calculate the angle θ to attain such centripetal force for the quadcopter. (2 marks)

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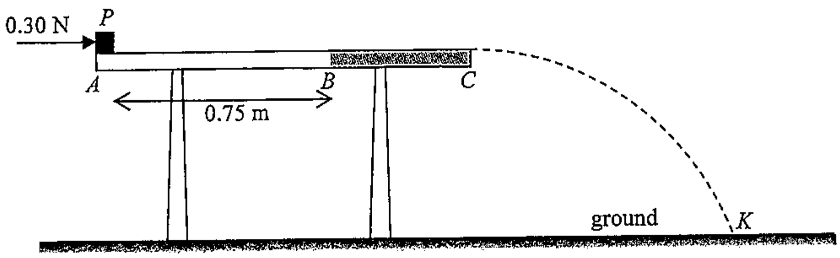
3. A small block P of mass 0.20 kg rests on side A of a horizontal table as shown. The surface of the table is smooth from A to B but is rough from B to C. Neglect air resistance.

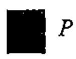
Figure 3.1



A horizontal force of 0.30 N acts on the block as shown.

(a) Find the speed of the block after it travels 0.75 m to point B. (2 marks)

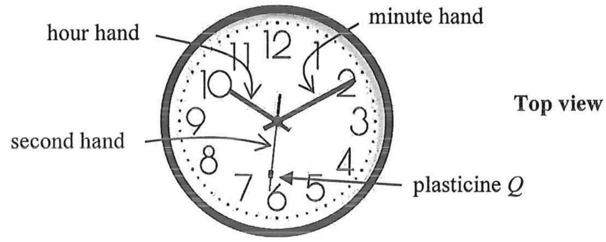
(b) The horizontal force is removed when block P just reaches B. When the block continues to travel from B to C, on the free-body diagram below indicate and name the force(s) acting on the block. (2 marks)



*(c) The block leaves side C of the table and takes 0.35 s to reach point K on the ground. Estimate the height of the table. (2 marks)

*4. A large clock facing upwards is placed horizontally on a table as shown. A small piece of plasticine Q rests on the steadily rotating second hand of the clock.

Figure 4.1

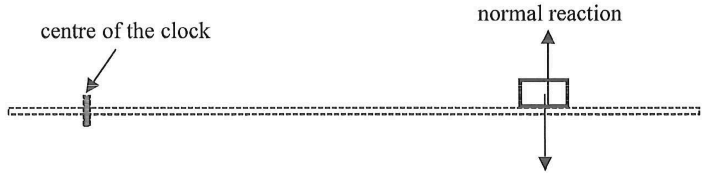


(a) Find the angular speed ω, in rad s^{-1}, of the second hand of the clock. (1 mark)

(b) Figure 4.2 shows the side view of the second hand of the clock.

Figure 4.2

In the free-body diagram of plasticine Q below, add a force acting on Q to complete the diagram and name this force. (1 mark)



(c) A student claims that the centripetal force required for Q will become smaller if it is placed nearer to the centre of the clock. Explain briefly whether this claim is correct or not. (2 marks)

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